

What makes models robust?

Margret Keuper



How reliable are Computer Vision Models?

- To be successful, ML Models (also Foundation Models) need to be reliable.
- How can we understand their behavior?

Example: Classification of
OOD images
ClimateTC Dataset



10.7% a polar bear.

0.01% an arctic fox.

89.2% a seal.

Radford et al. (2021). Learning Transferable Visual Models From Natural Language Supervision. *International conference on machine learning*. PMLR.

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Example of
OOB
Similar problems as most previous deep learning models!



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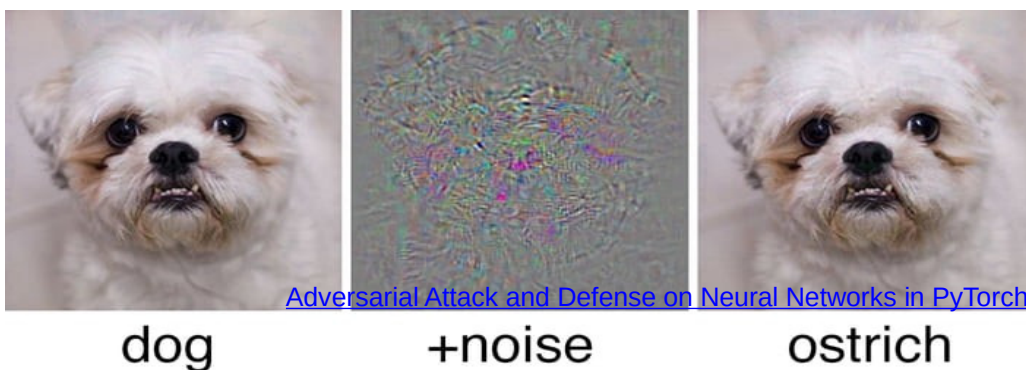
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Computer Vision Models - Robustness

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 - **Robust Deep Learning for Computer Vision?**



Current models can be easily fooled by adversarial attacks.

snow

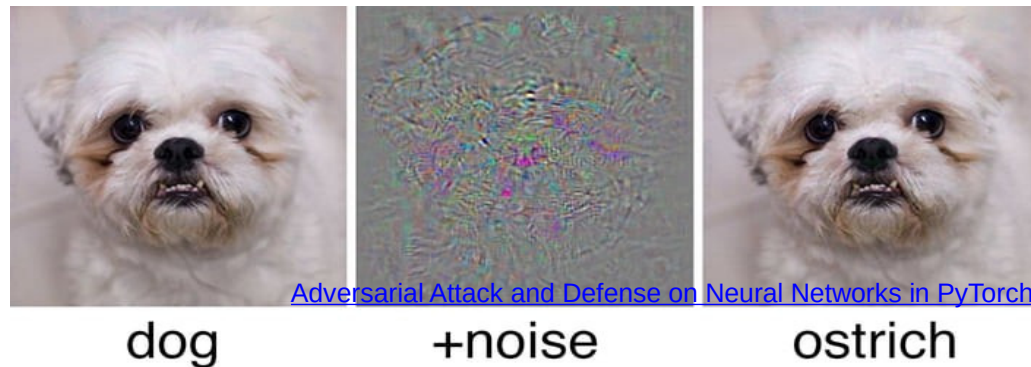
sand



Current models don't generalize well to unseen domains (ACDC dataset)

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Current models can be easily fooled by adversarial attacks. → Reliability

snow



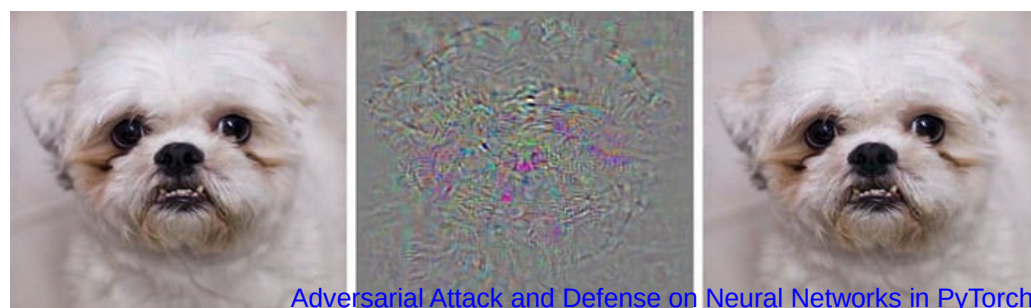
sand



Current models don't generalize well to unseen domains (ACDC dataset) → Generalization Ability

Computer Vision Models - Robustness

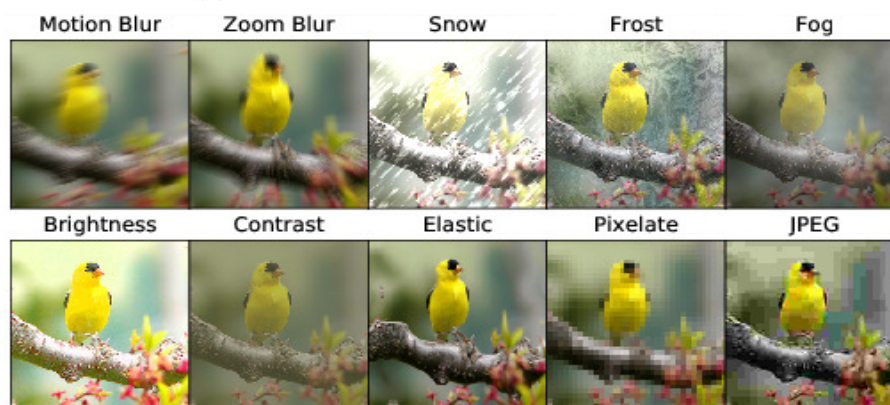
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dog

+noise

ostrich



ImageNet-C, Hendrycks et al.,

ICLR'19

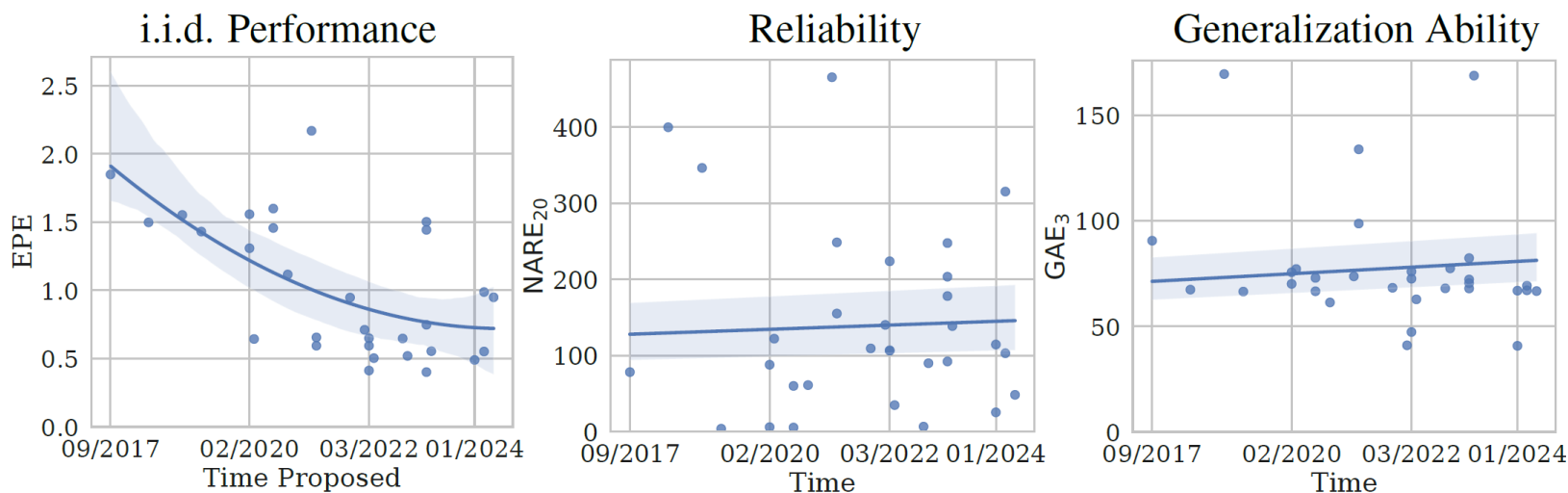
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Current models can be easily fooled by adversarial attacks. → Reliability

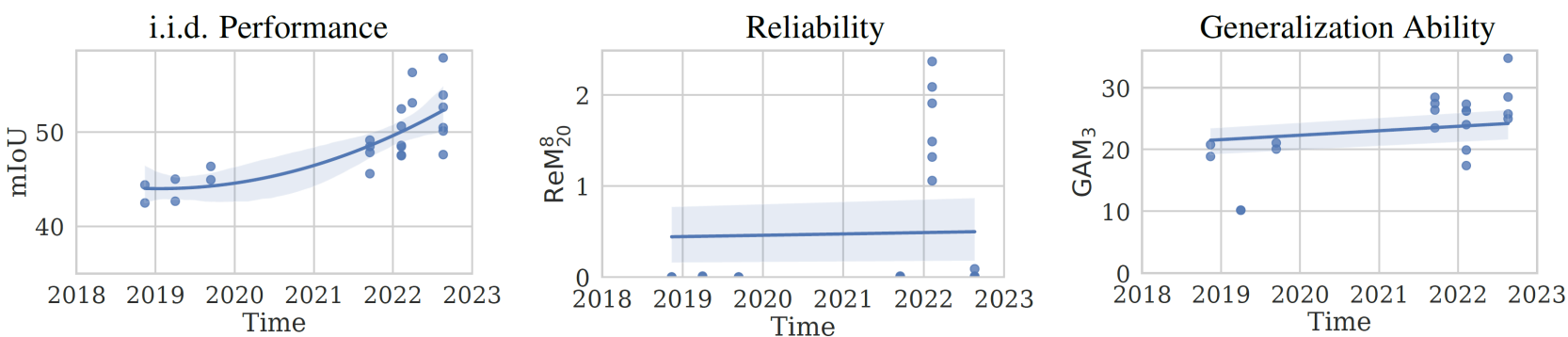
Current models don't generalize well to unseen domains
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How reliable are Computer Vision Models?

Optical Flow
Estimation
Methods



Semantic
Segmentation
Methods

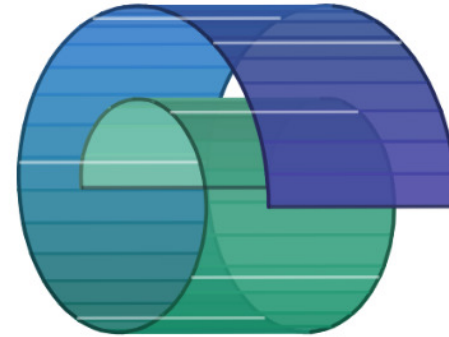


[FlowBench: Benchmarking Optical Flow Estimation Methods for Reliability and Generalization](#), S. Agnihotri, J. Y. Caspary, L. Schwarz, J. Schmalfluss, A. Bruhn, M. Keuper, Transactions on Machine Learning Research (TMLR) 2025.

What makes models robust?

Hypothesis: The lack of robustness in NNs has **multiple causes**

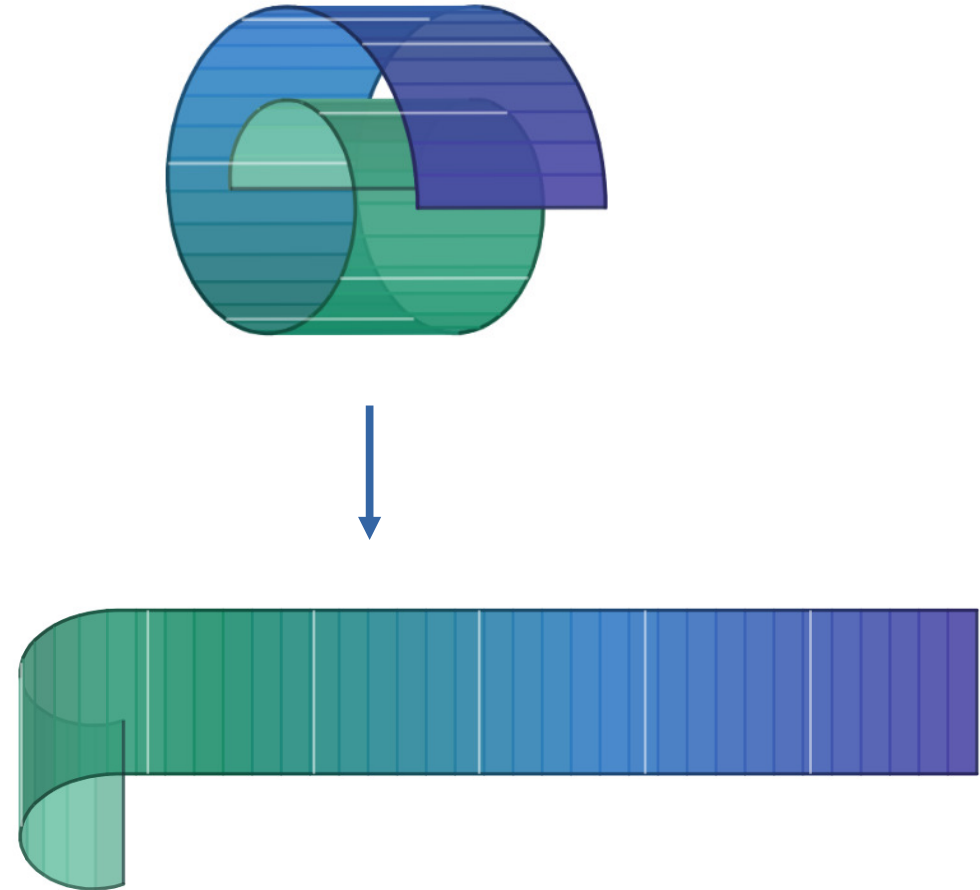
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 - hard to analyze (Kiani et al., Neurips 2024)



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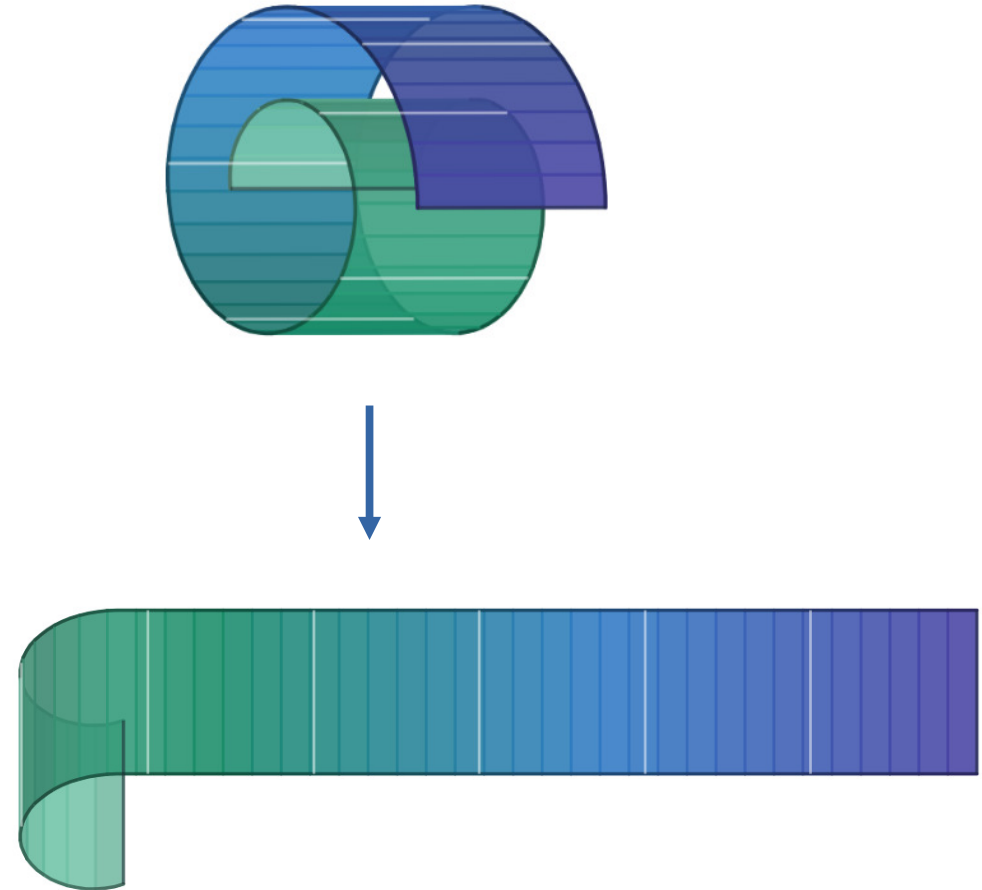
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 - Loss
 - Data Augmentation

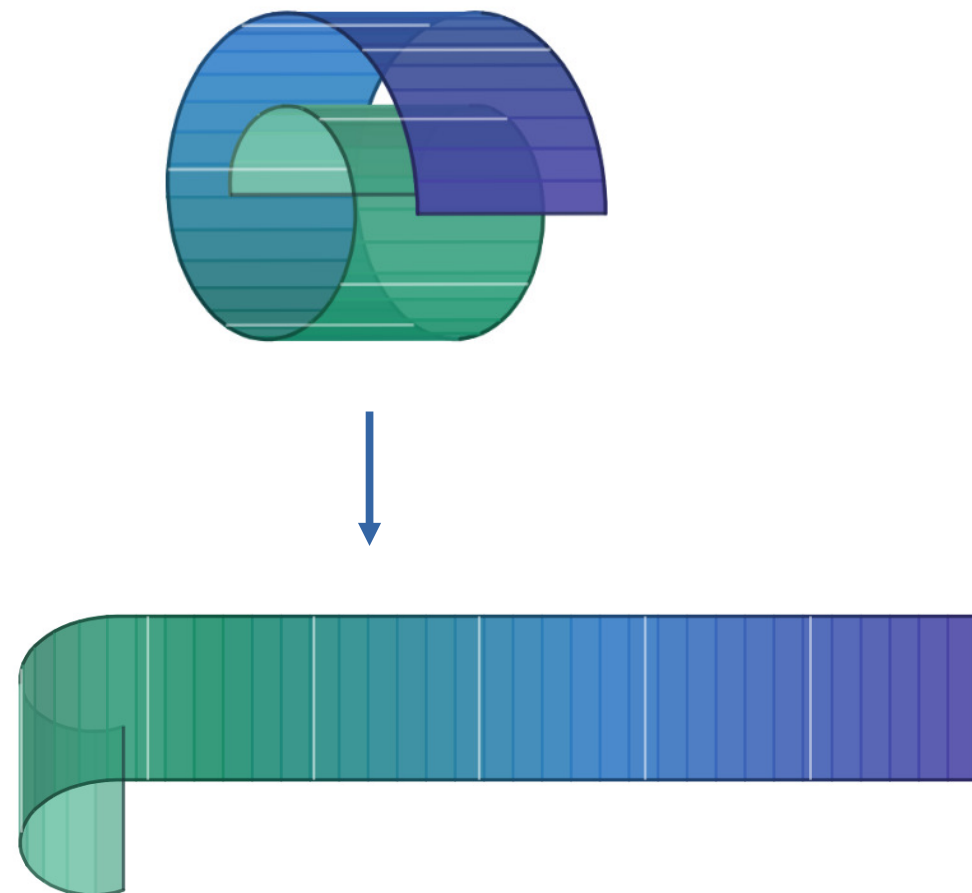


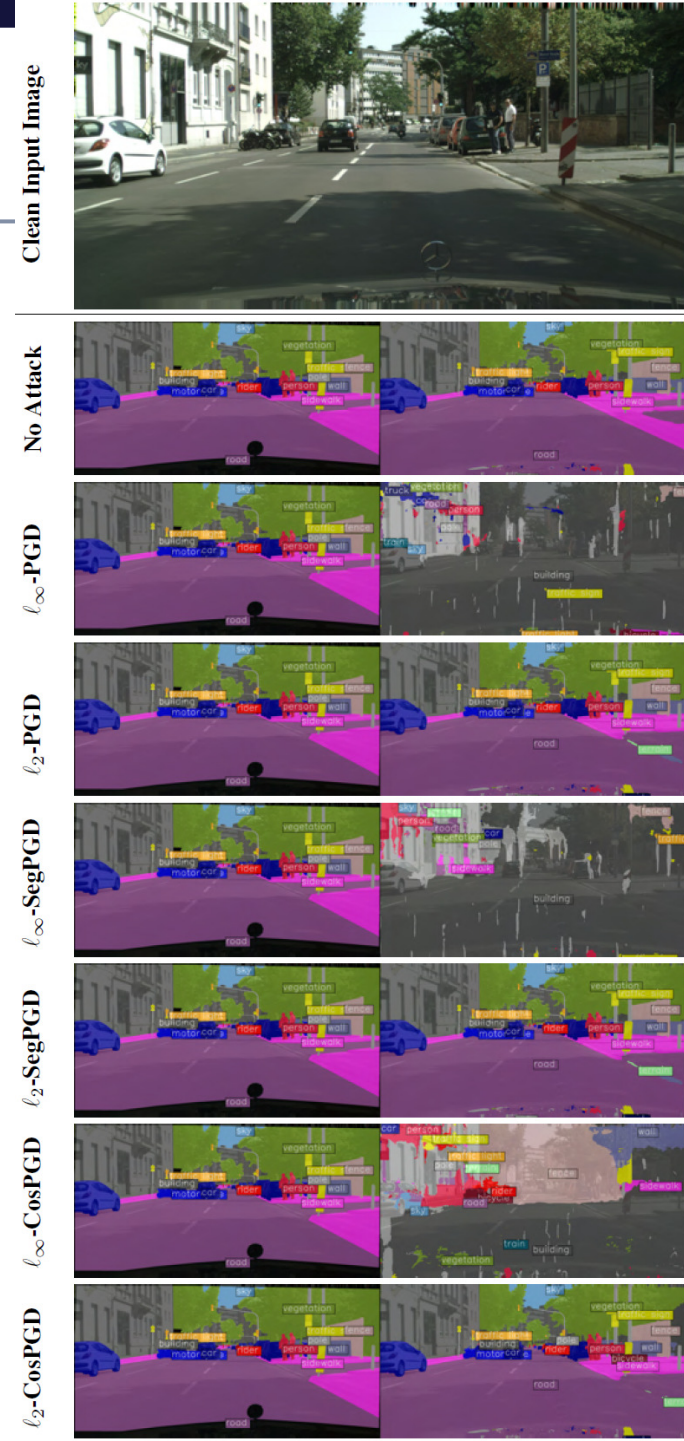
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What can we learn from analyzing trained models for pixel-wise tasks?





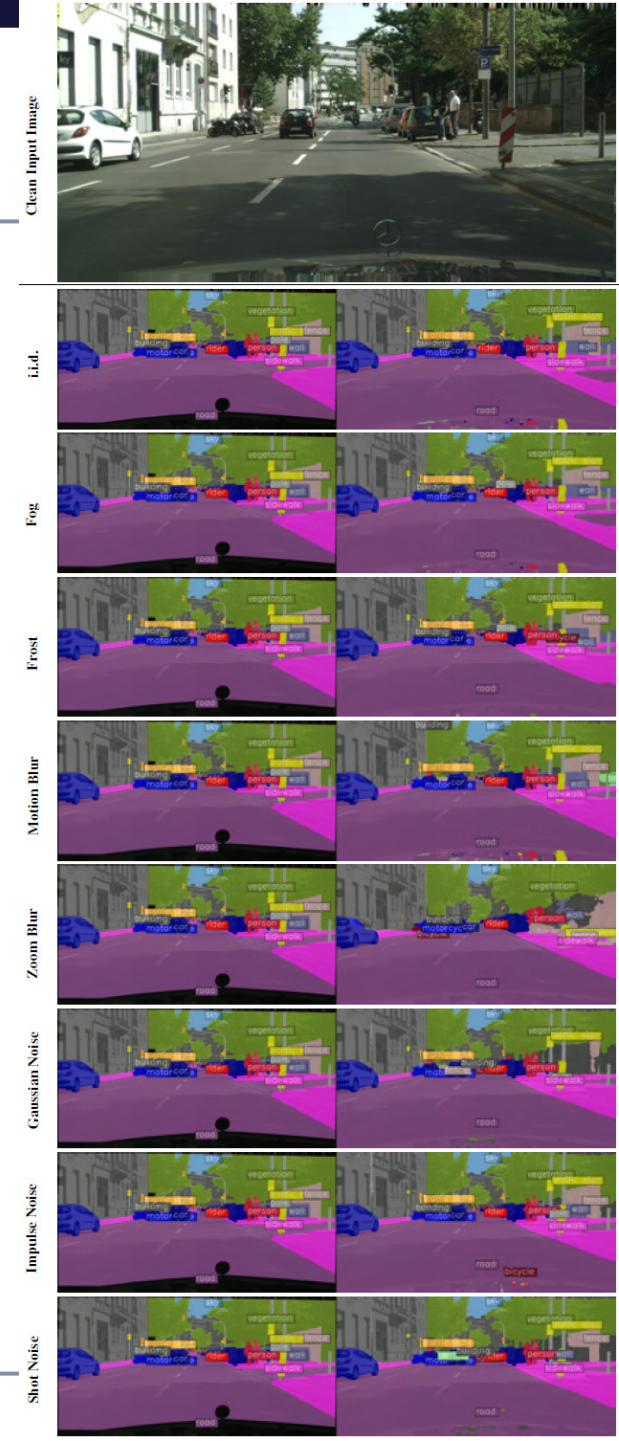
Robustness Benchmarking

← For Reliability Measure

For Generalization Ability Measure →

SEMSEGBENCH & DETECBENCH: Benchmarking Reliability and Generalization Beyond Classification, S. Agnihotri et al, arXiv 2025

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Robustness Benchmarking

1. High Positive Correlation in performance measured using different metrics (Segmentation)

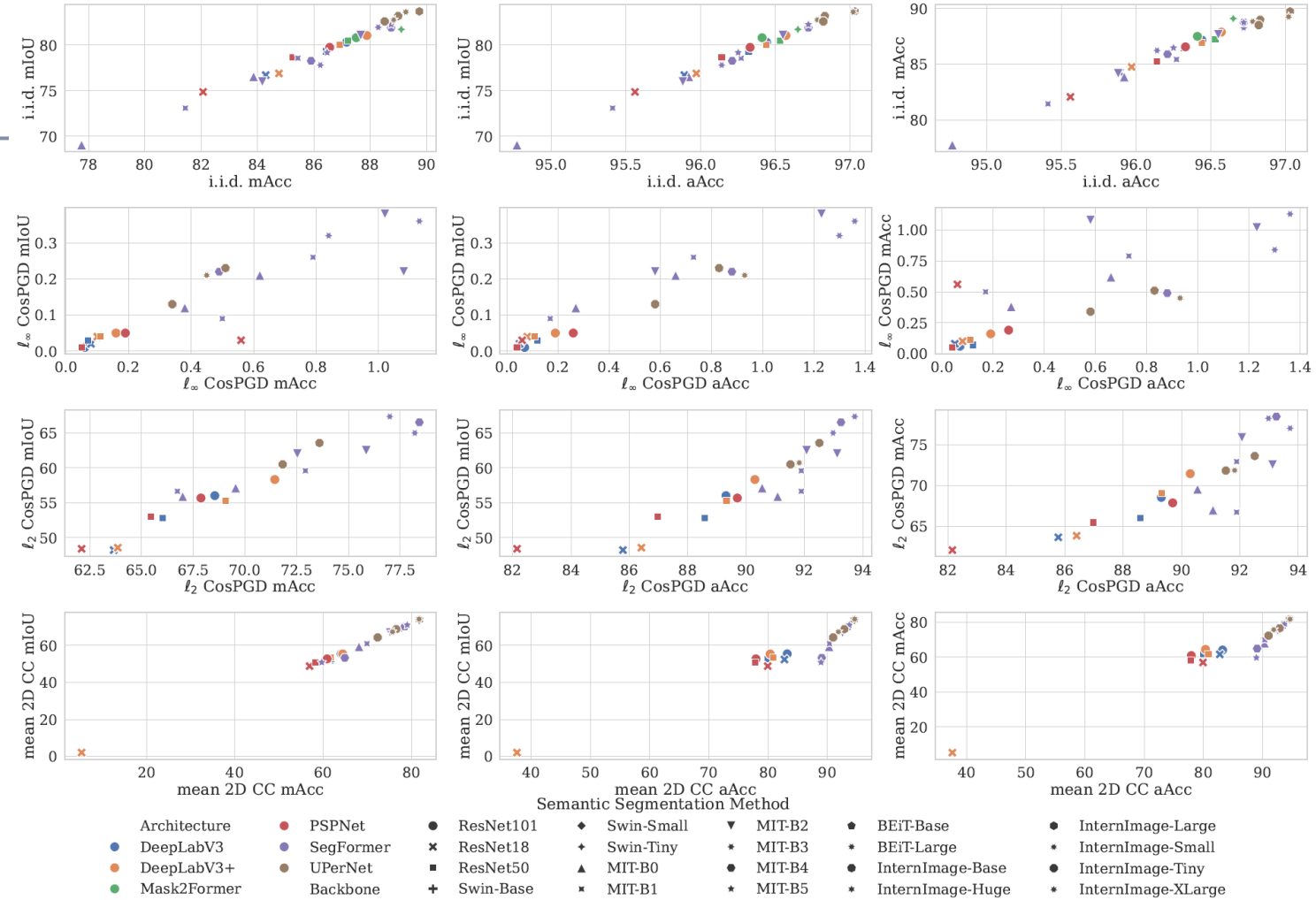
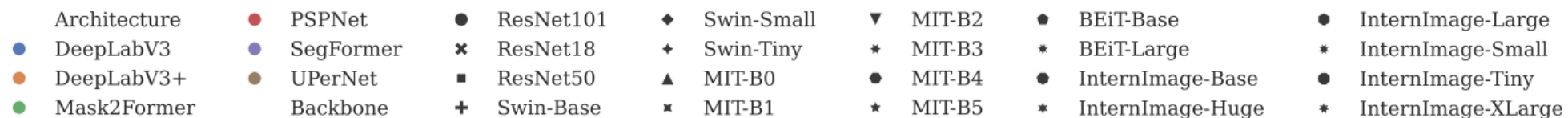
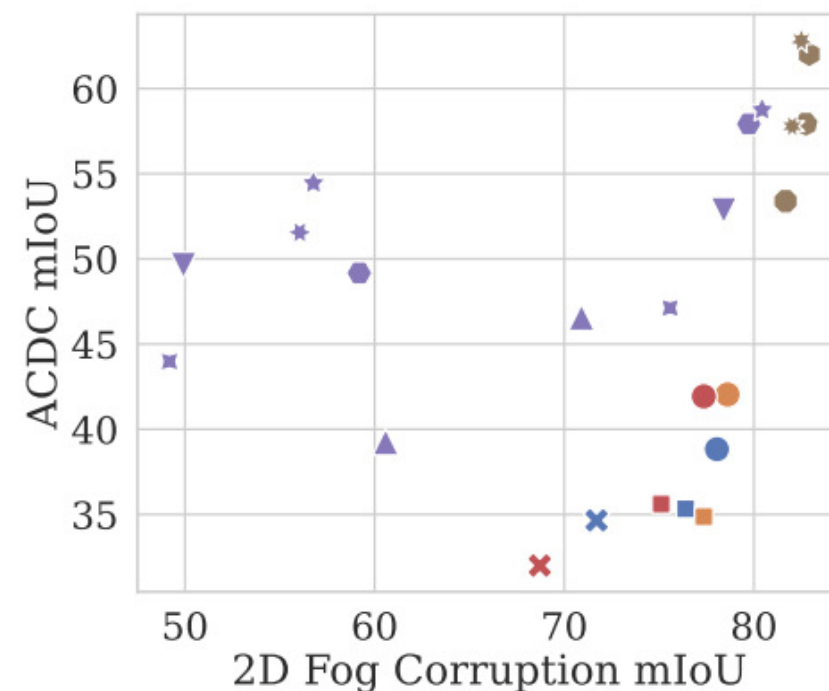
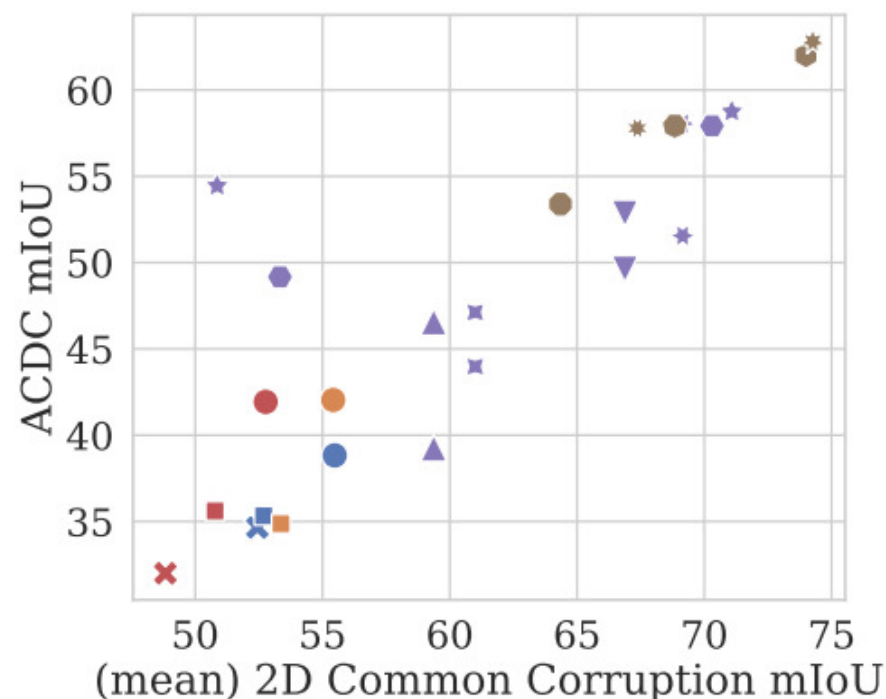


Figure 8. **Dataset used: Cityscapes.** The performance of semantic segmentation methods is usually captured across different metrics, namely: mIoU, mAcc, and aAcc. In the analysis of this work, we only used mIoU. Here we show that there is a high positive correlation between these metrics, and observations made using mIoU would still hold using other metrics. The top row is correlation when using i.i.d. data for evaluations. The second row is when using ℓ_∞ -norm bounded CosPGD attack with $\epsilon = \frac{8}{255}$ and $\alpha=0.01$. The third row is when using ℓ_2 -norm bounded CosPGD attack with $\epsilon = 64$ and $\alpha=0.1$. The fourth row is when using 2D Common Corruptions, here we calculate the mean for each metric across all 2D Common Corruptions. Colors are used to show different architectures and marker styles are used to show different backbones used by the semantic segmentation methods.

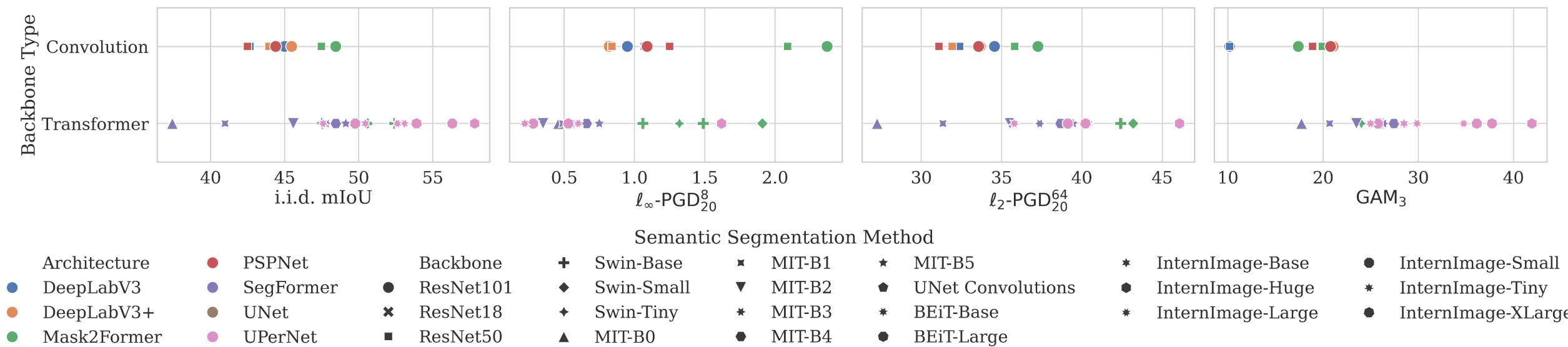
Robustness Benchmarking – Semantic Segmentation

High Positive Correlation between Real-World Corruptions and Synthetic Corruptions



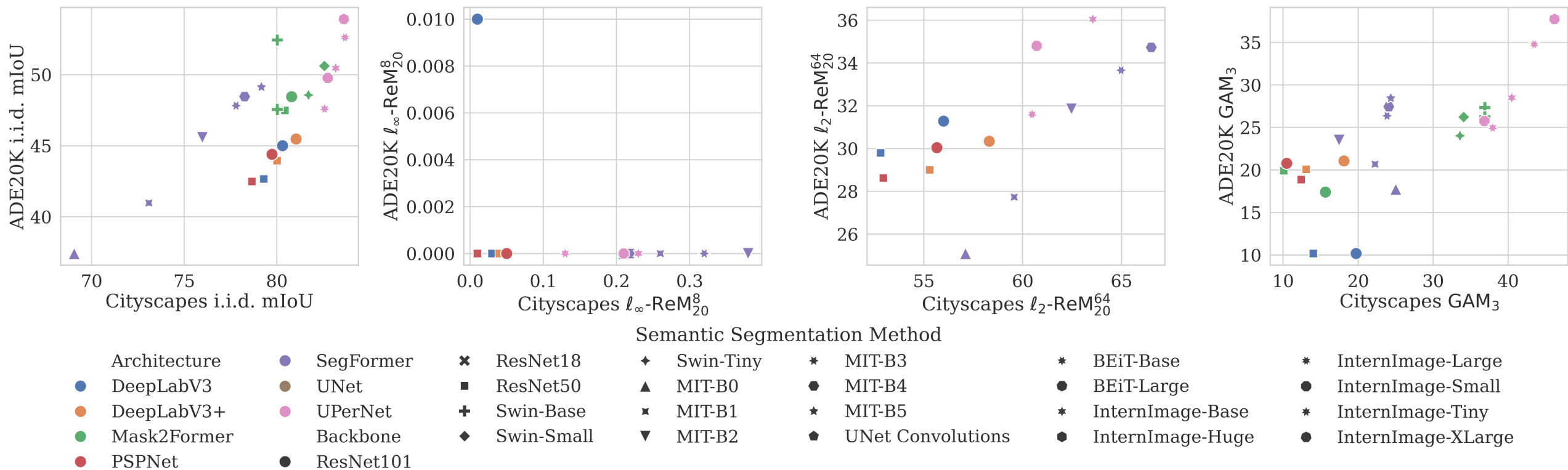
Robustness Benchmarking – Semantic Segmentation

Influence of Backbone (ADE20K): Transformer show better OOD Robustness (in line with expectations from image classification)



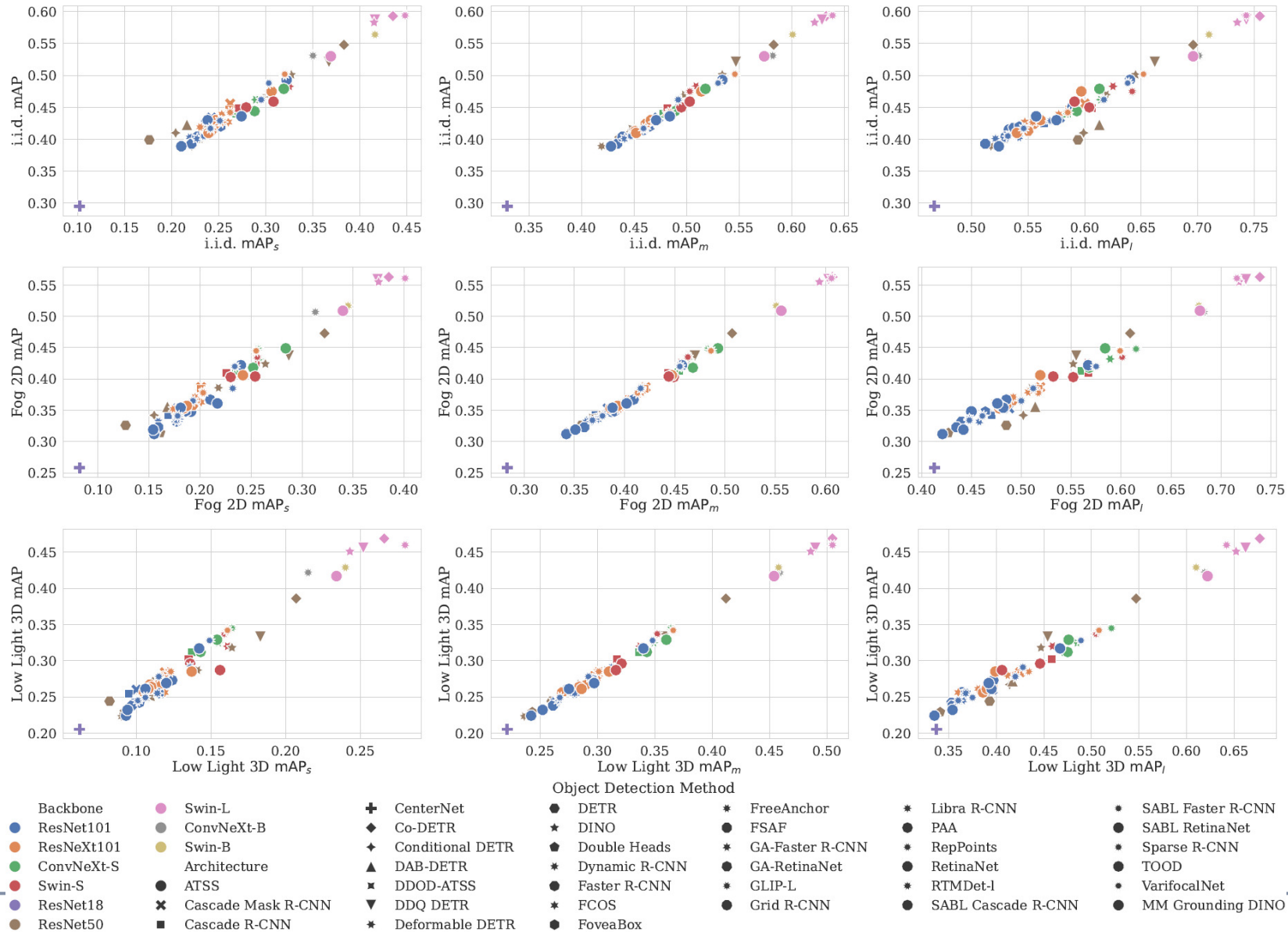
Robustness Benchmarking – Semantic Segmentation

Some Positive Correlation between Performance on different datasets



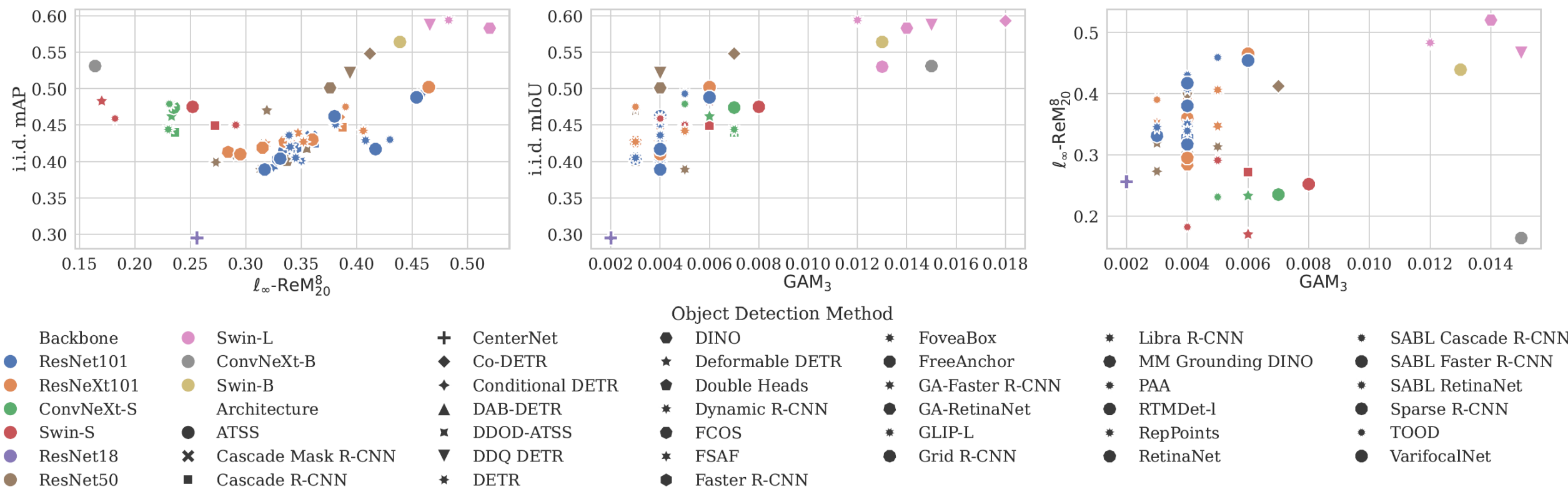
Robustness Benchmarking – Object Detection

1. High Positive Correlation between different metrics



Robustness Benchmarking – Object Detection

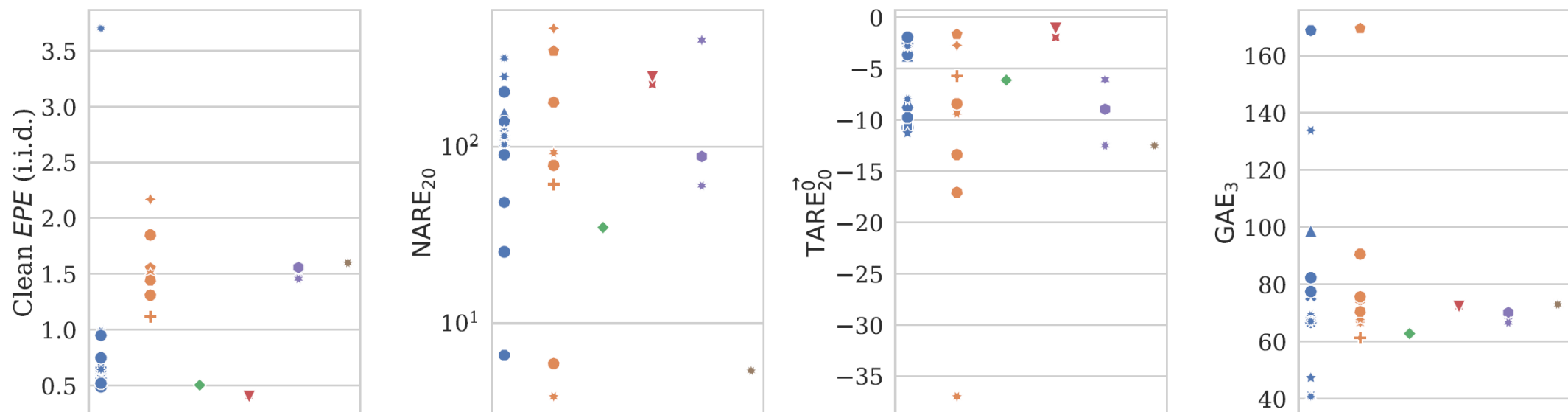
i.i.d. Performance v/s Reliability v/s Generalization Measure: CORRELATION



Robustness Benchmarking – Optical Flow

Method Families

Transformer based methods are not more robust --> Are we using sufficient training samples?



Method With Family

Family	Model	Method With Family	Method With Family	Method With Family	Method With Family
● RAFT	● CCMR	▲ Flow1D	* IRR	* MaskFlowNetS	* RPKNet
● PWC	✕ CRAFT	✕ FlowFormer	● LLA-Flow	● MaskFlowNet	● SKFlow
● Deep Inverse Patchmatch	■ CSFlow	▼ FlowFormer++	* LiteFlowNet	* NeuFlow	* STaRFlow
● FlowFormer	+ DICL-Flow	* GMA	● LiteFlowNet2	● PWCNet	● ScopeFlow
● FlowNet	◆ DIP	● GMFlow	* LiteFlowNet3	* RAFT	* SplatFlow
● SSRFlow	✦ FastFlowNet	* GMFlowNet	● MS-RAFT+	● RAPIDFlow	● VideoFlow
		★ HD3			

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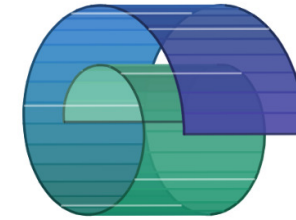
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Classical rules such as: “bigger models are more robust” or “self-attention helps” don’t always apply.

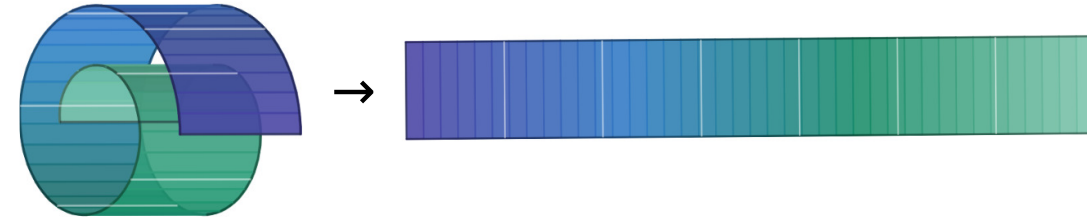
→ Impact of Loss and Data!

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Can we directly simplify the manifold?

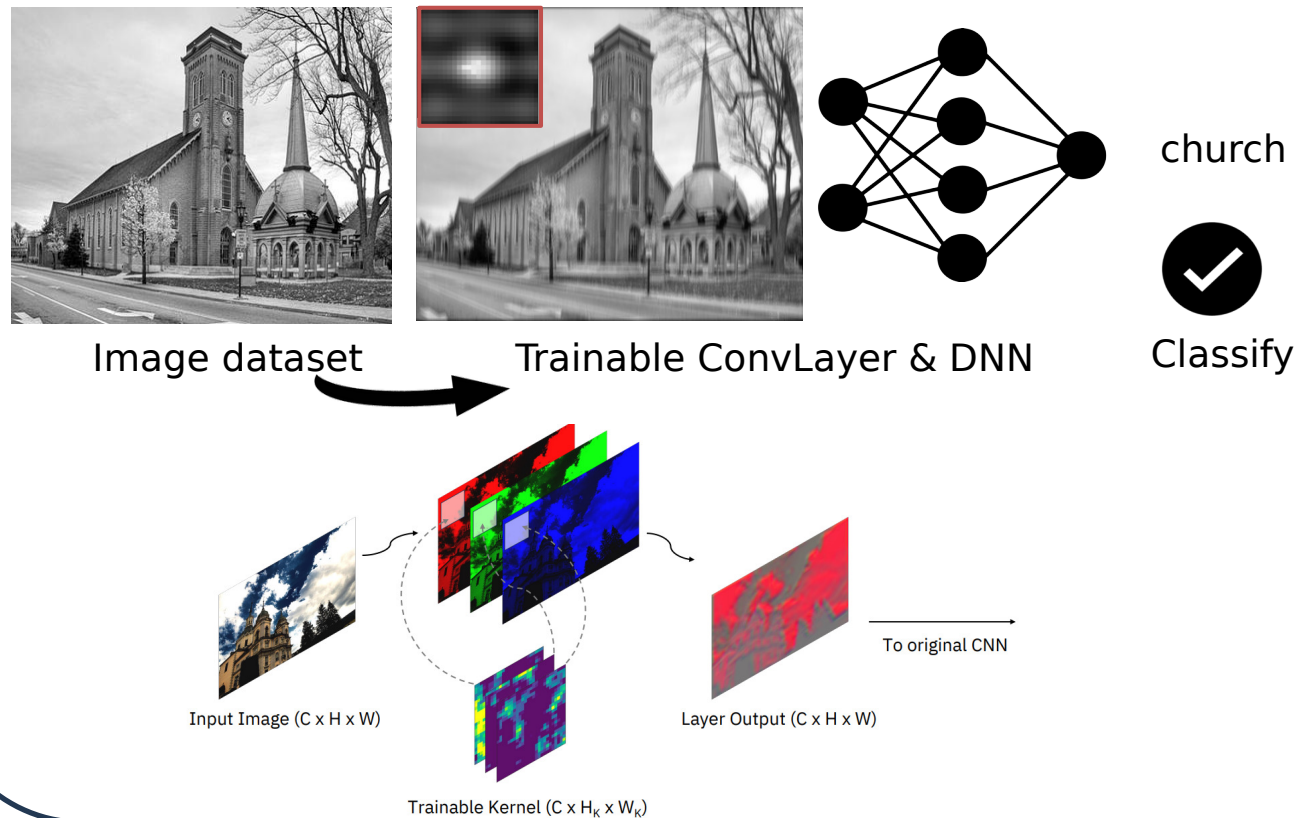


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→ Impact of Loss and Data!

What makes models more robust?

Over-completeness in learned representations might encourage overly complicated manifolds

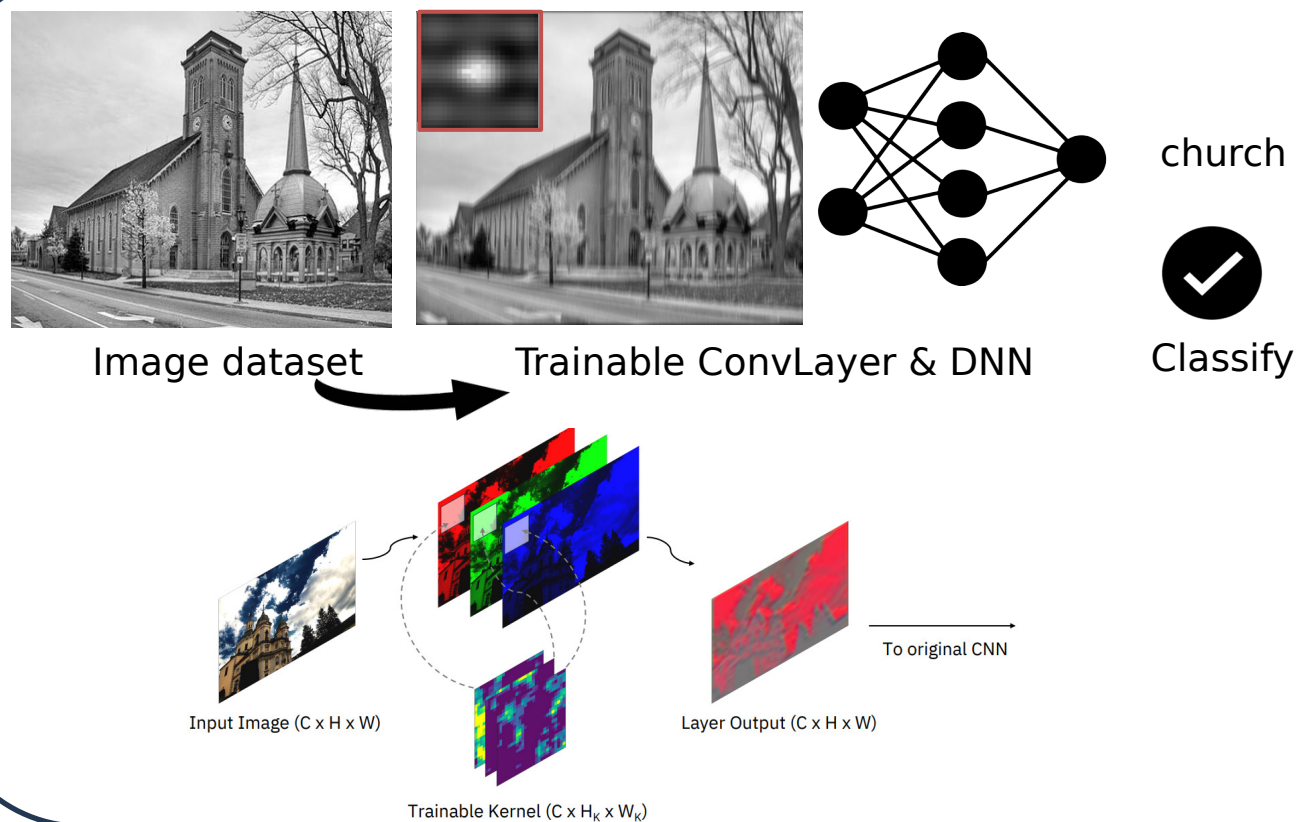


Learning the Essential in Less Than 2k Additional Weights (2024)

K. Bäuerle, P. Mueller, S. Kazim,
I. Ihrke, M. Keuper



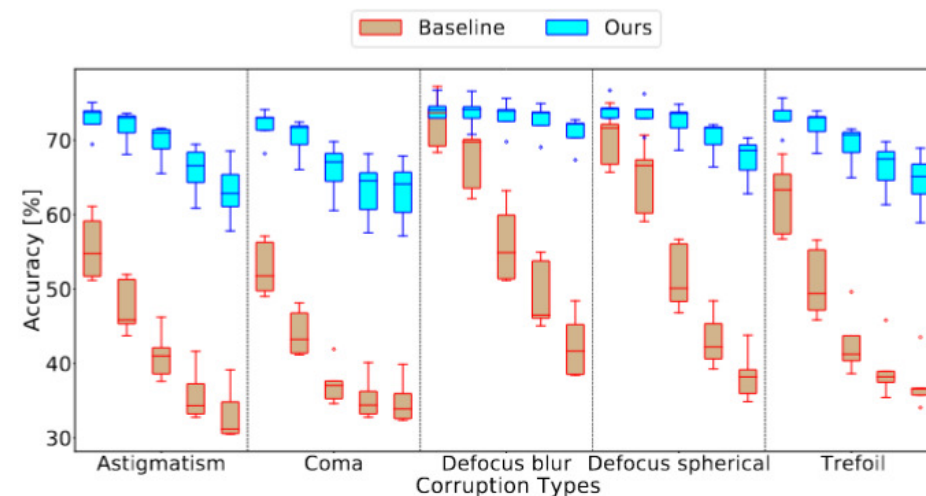
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Learning the Essential in Less Than 2k Additional Weights (2024)

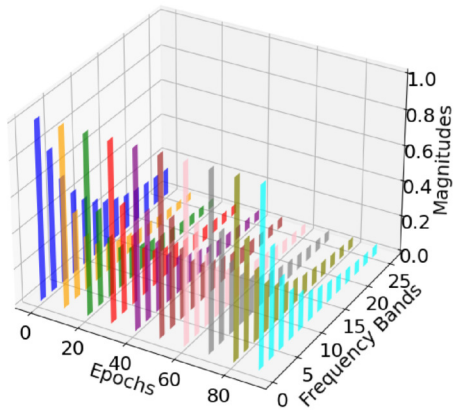
K. Bäuerle, P. Mueller, S. Kazim, I. Ihrke, M. Keuper

Result in OpticsBench ImageNette with ResNet 50

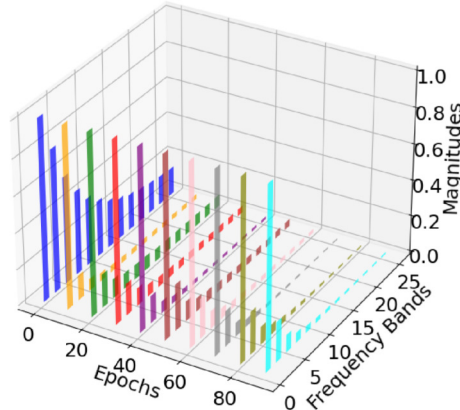


What makes models more robust?

What kernels are learned?
On ImageNette (10 classes)



No regularization



L1 regularization

class I: content preserving (full rank)
class II: trainable (potentially full rank)
class III: projection (low rank)

Kernel type	class	CD	OB	CC
None (Base)	-	*0.800	*0.592	*0.487
Preserve content	I	0.754	0.476	0.513
Preserve content large	I	0.645	0.478	0.440
Conv2D KS=25	II	0.655	0.601	0.501
Fully trainable	II	<u>*0.775</u>	<u>*0.685</u>	<u>*0.565</u>
Random initialization	II	0.711	0.673	0.550
L1 prior	III	0.712	0.699	0.567
Directional blur filter	II	0.768	0.437	0.345
Gauss blur filter	II	0.764	0.668	0.541

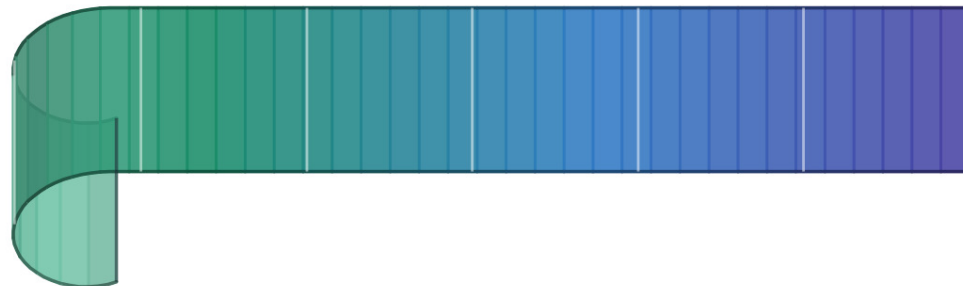
What makes models robust?

large model → highly expressive
→ need for many training examples



Still, the resulting representation is usually over-complete, some dimensions of the representation are not well defined.

However, when the model faces an initial bottleneck – it can even simplify the representation on its own!



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Test: For **more difficult tasks on the same capacity**, would it still work?

What makes models more robust?

... as the task becomes harder (and capacity remains the same)

→ the gap becomes smaller

ImageNet-100

	clean	OpticsBench	Common Corruptions
Baseline ResNet 50	80.1%	53.6%	40.6%
trainable projection	79.7%	55.8%	43.7%

ImageNet-1k

	clean	OpticsBench	Common Corruptions
Baseline ResNet 50	78.1%	48.2%	39.3%
trainable projection	77.4%	50.9%	42.5%

What makes models robust?

large model → highly expressive
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Still, the resulting representation is usually over-complete, some dimensions of the representation are not well defined.

Self-regularization is limited, at least when conflicting with accuracy.

What makes models more robust?

What can we understand from basic signal processing?

- FrequencyLowCut Pooling -- Plug & Play against Catastrophic Overfitting, J. Grabinski, S. Jung, J. Keuper, M. Keuper, ECCV 2022.
- Fix your downsampling ASAP! Be natively more robust via Aliasing and Spectral Artifact free Pooling, J. Grabinski, S. Jung, J. Keuper, M. Keuper, IJCV 2026.
- Improving Stability during Upsampling-- Spectral Artifacts and the Importance of Spatial Context, S. Agnihotri, J. Grabinski, M. Keuper, ECCV 2024.

Towards more robust CNNs ...

CNNs apply a sequence of convolution and downsampling operations.

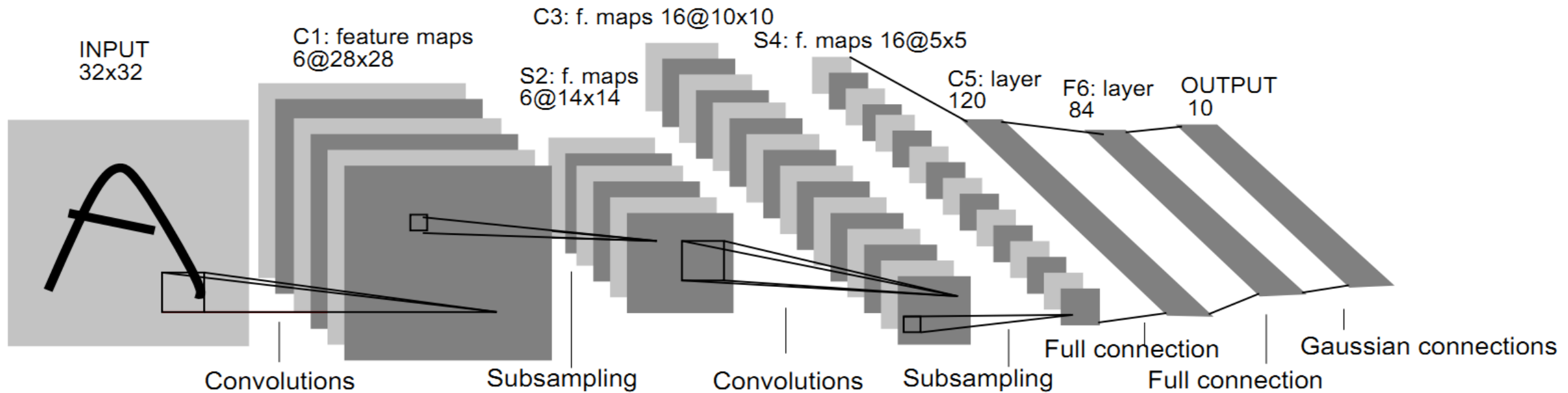


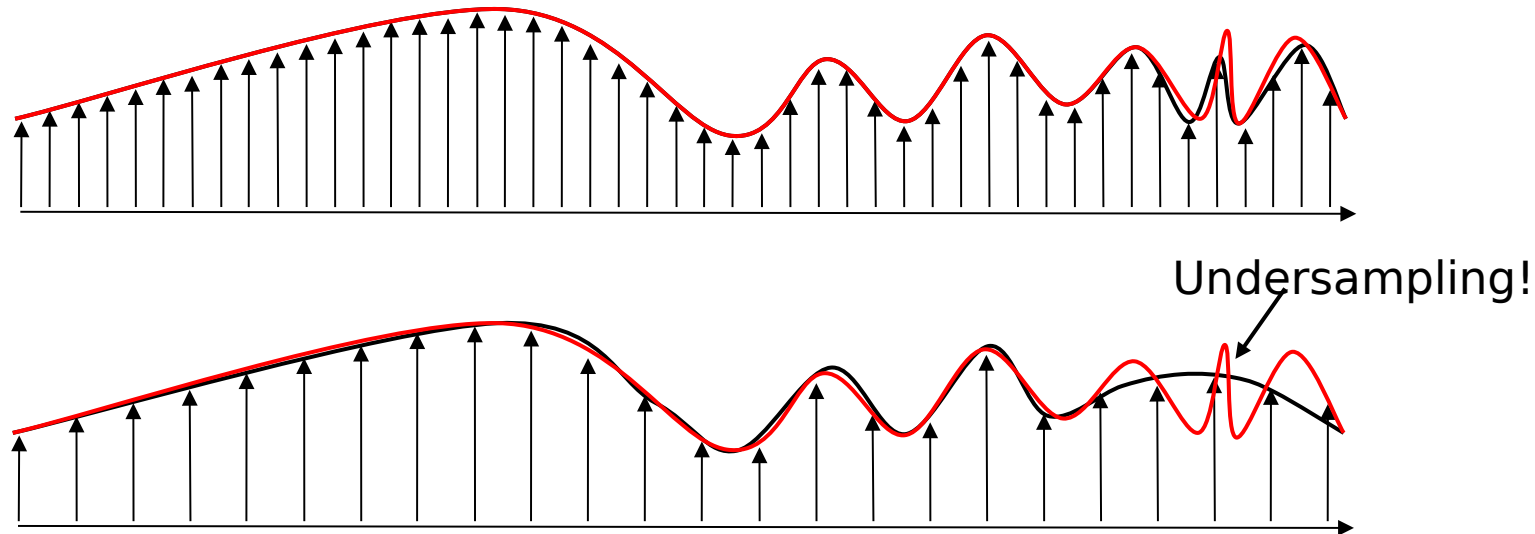
Fig. 2. Architecture of LeNet-5, a Convolutional Neural Network, here for digits recognition. Each plane is a feature map, i.e. a set of units whose weights are constrained to be identical.

LeNet: Proposed by LeCun et al. in 1998.

Downsampling in most CNNs ...

What can go wrong during sampling?

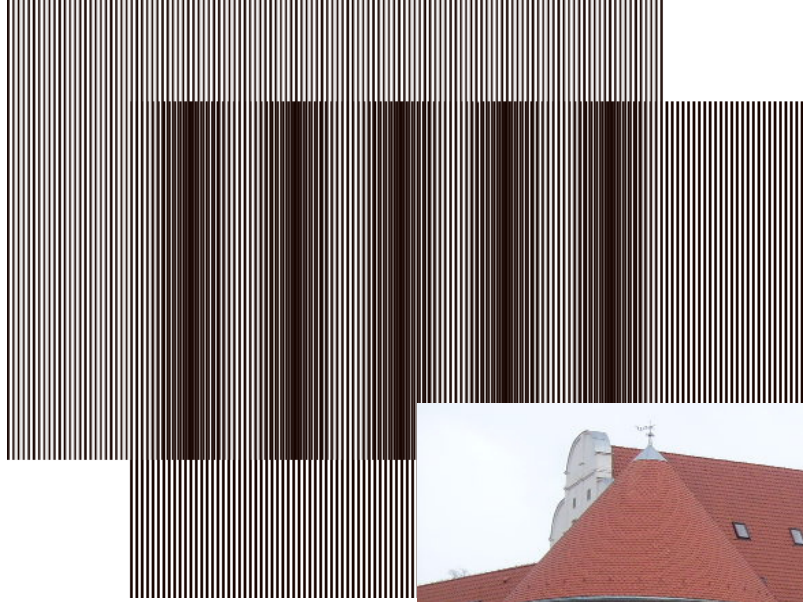
Signal
Reconstructed
signal



Ambiguities! - High frequencies are confused with low frequencies.

Downsampling in most CNNs ...

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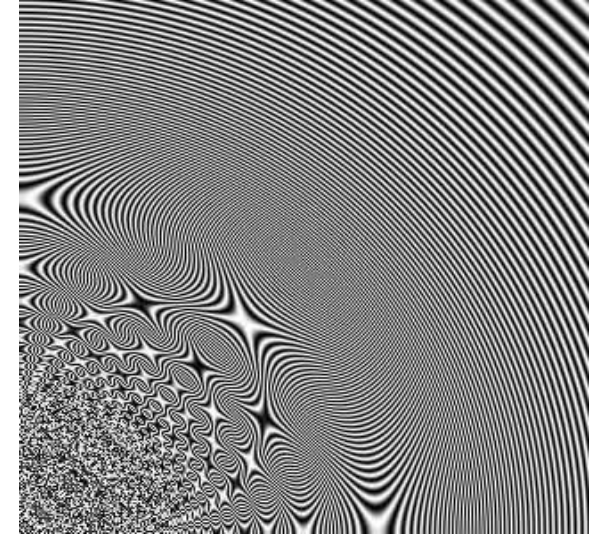


Moirée Pattern:
Aliasing durch
Überlagerung
von Texturen



A: CC BY-SA 3.0,
<https://commons.wikimedia.org/w/index.php?curid=192995>

B: Von Colin Pelka -
Eigenes Werk
(Originaltext: eigene
Arbeit), Gemeinfrei,
<https://commons.wikimedia.org/w/index.php?curid=19614895>



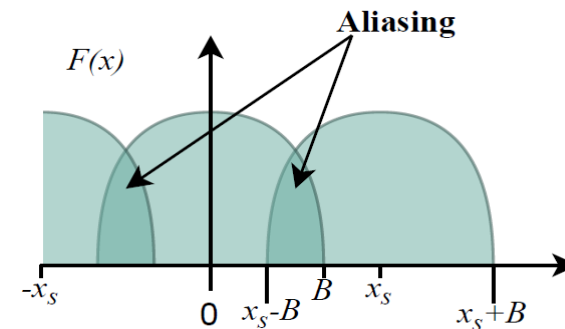
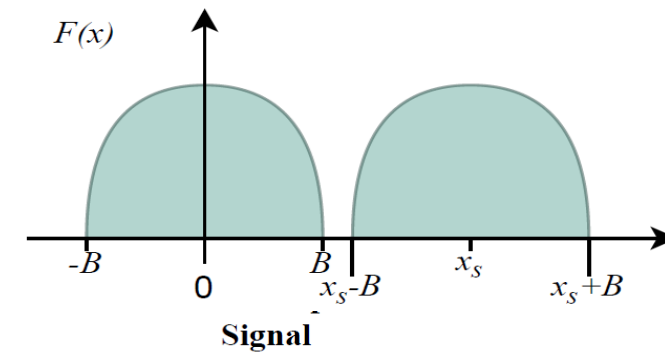
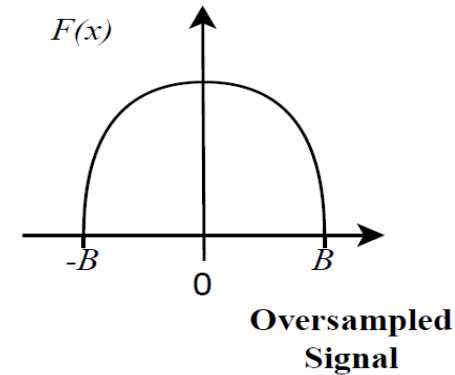
Gamito und Maddoc (2006)
[Anti-aliasing with Stratified B-spline Filters of Arbitrary Order](#)

Frequency Domain

In the Fourier domain, sampling causes a replication of the signal, where the distance of replica is proportional to the sampling rate.

Oversampling – signal can be reconstructed

Undersampling – signal suffers from aliasing



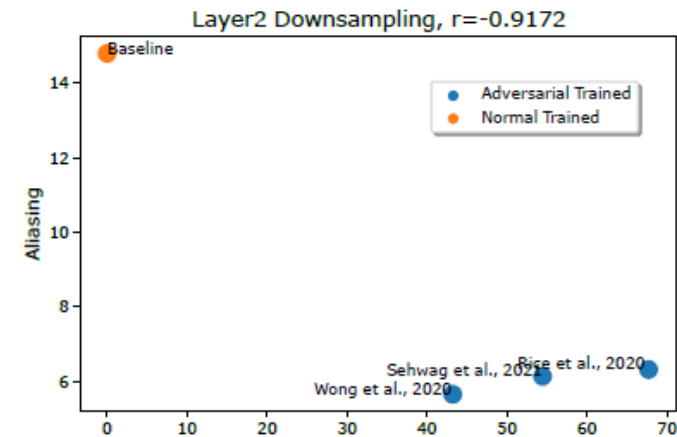
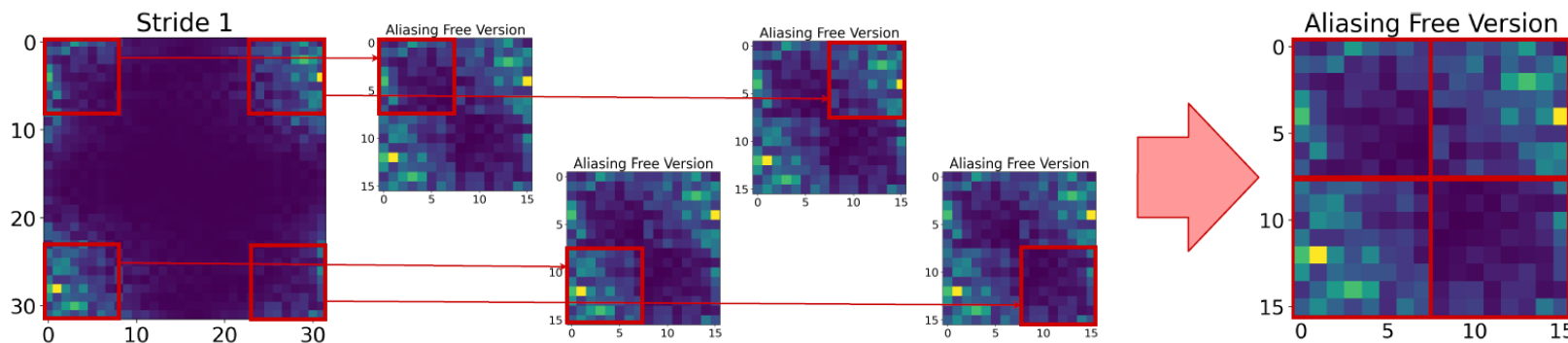
Frequency Domain – Analysis of Robustness

Aliasing and Adversarial Robust CNNs J. Grabinski, J. Keuper, M. Keuper, Springer Machine Learning, 2023.

Idea:

Estimate the **amount of aliasing after each downsampling** within the network – Compare Robust and non-robust models

Obtain the “**ground truth**” against which the standard downsampling can be compared to by sampling in the Fourier domain.



- Evaluate robustness against AutoAttack
- Robust models on RobustBench
- (Croce et al.'20)
- Train non-robust models without adv. training or add. data.

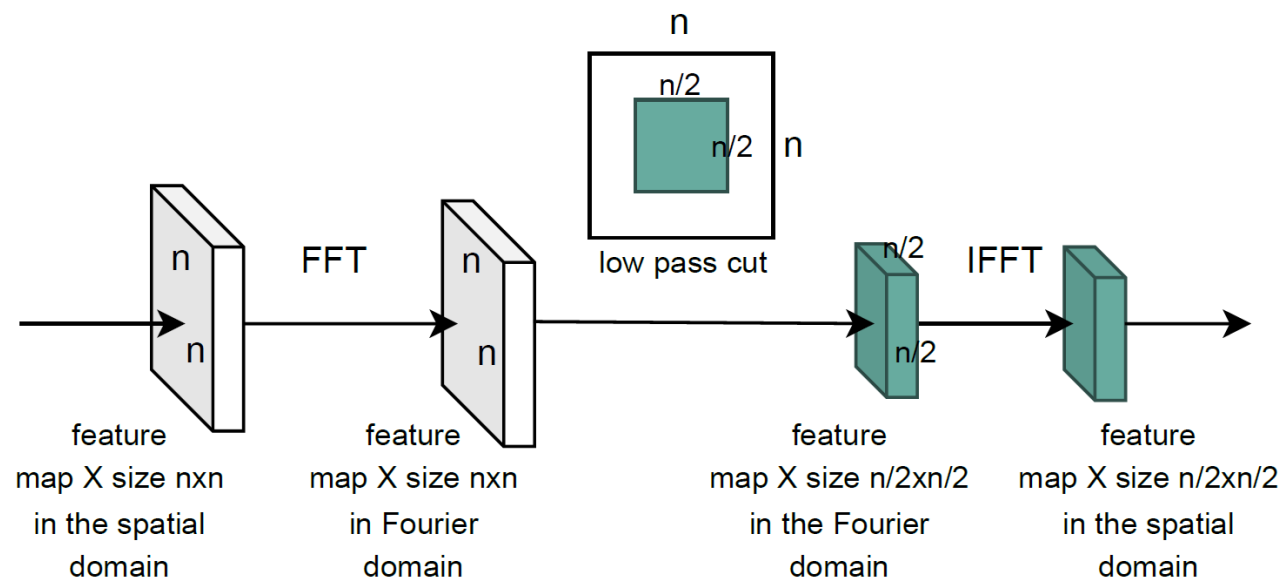
FrequencyLowCut Pooling

FrequencyLowCut Pooling – Plug&Play against Catastrophic Overfitting, J. Grabinski, S. Jung J. Keuper, M. Keuper, ECCV'22.

Transform each featuremap to the **Fourier domain**

Cut out the low frequencies only
(downsampling)

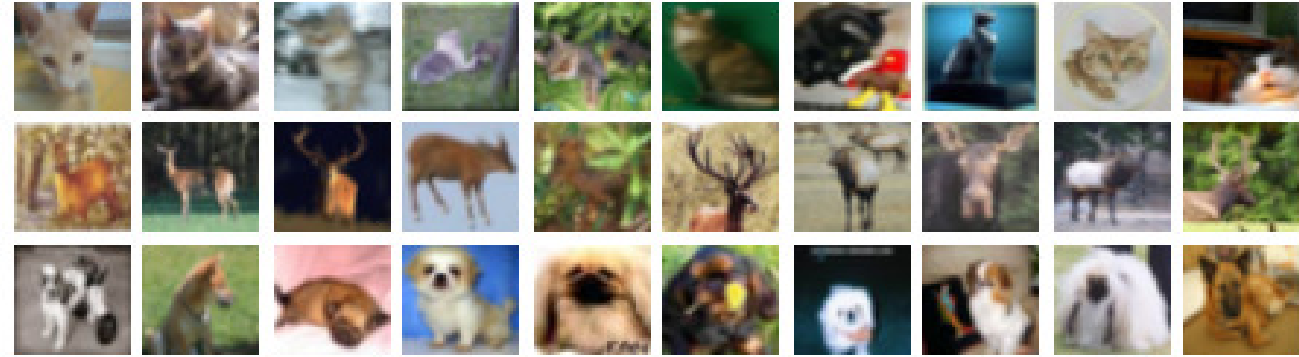
Transform back to the spatial domain



FrequencyLowCut Pooling

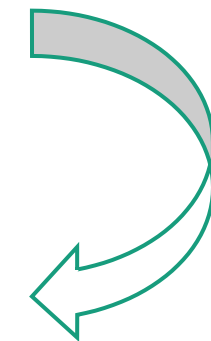
Results on Cifar10 – low resolution images

Many methods evaluated

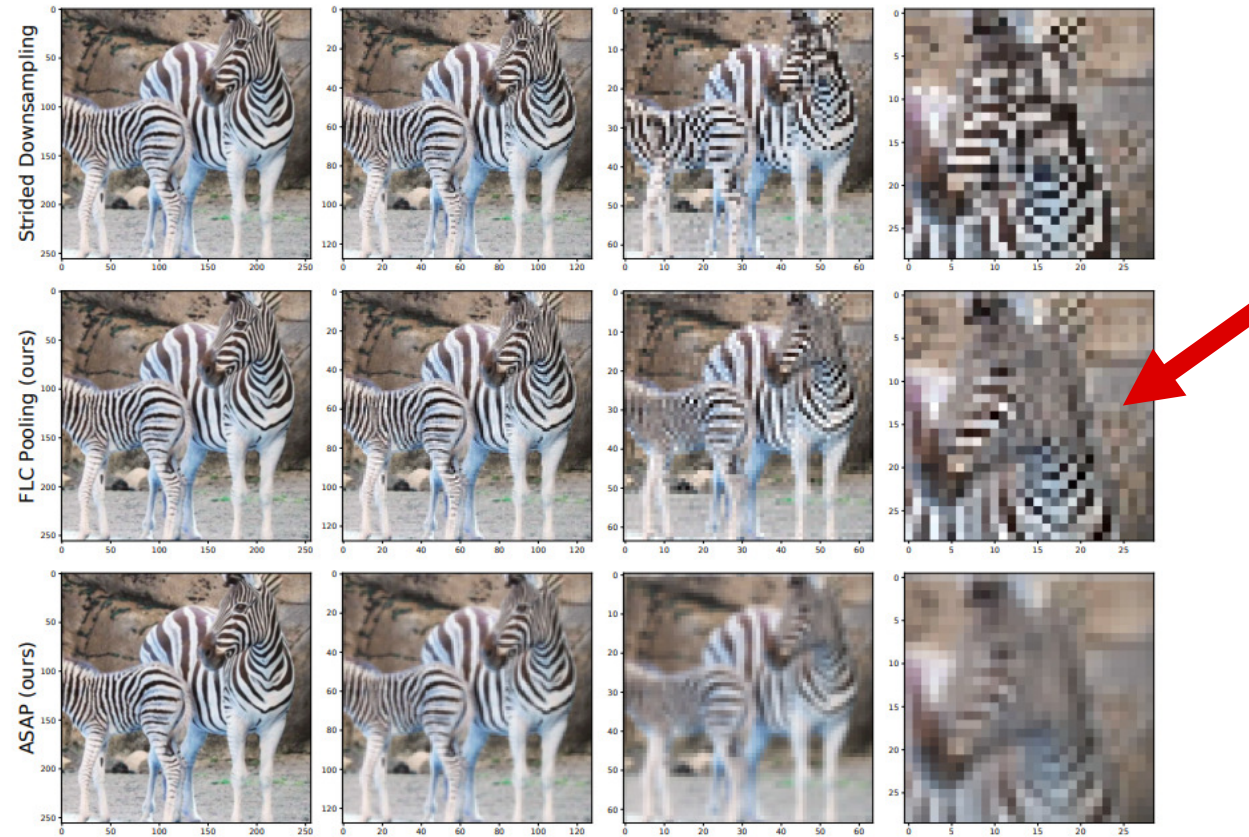


Method	Seconds per epoch (avg)	Clean Acc	AA Acc
Baseline	14.6 ± 0.1	95.08	0.00
FGSM & early stopping [43]	27.3 ± 0.1	82.88	11.82
FGSM & FLC Pooling (Ours)	34.5 ± 0.1	84.81	38.41
PGD [26]	115.4 ± 0.2	83.11	40.35
Robustness lib [11]	117 ± 19.0	76.37	32.10
TRADES [46]	219.4 ± 0.5	81.49	46.91
MART [42]	180.4 ± 0.8	55.49	8.63
AWP [44]	179.4 ± 0.4	82.61	49.43

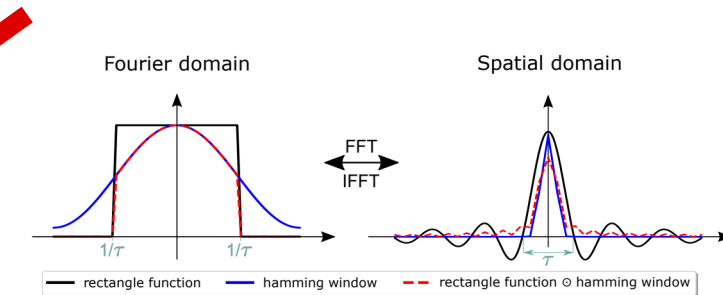
4 to 8
times
longer



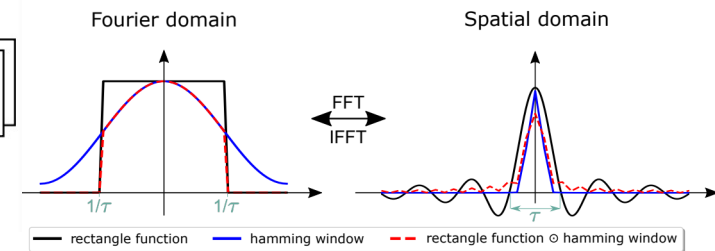
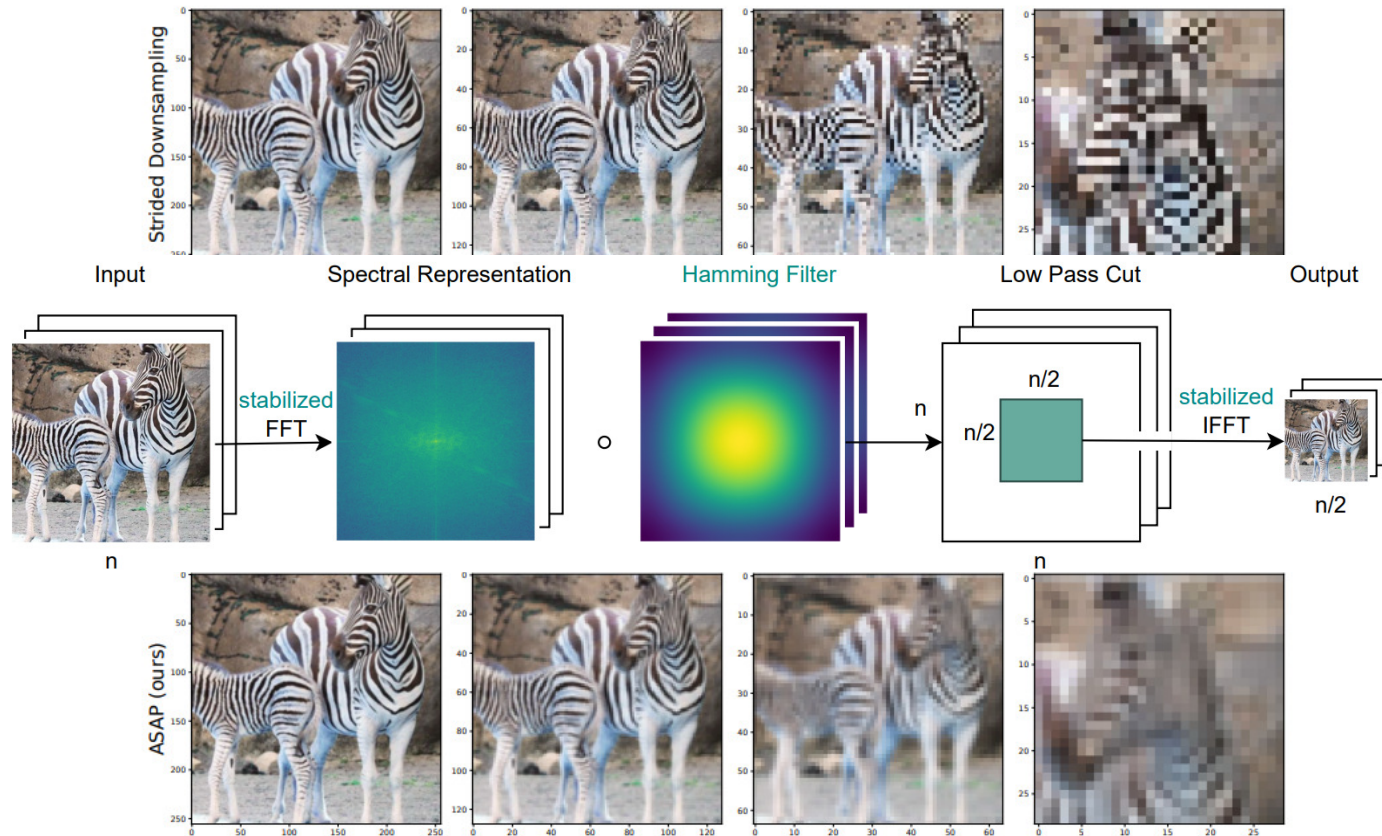
FrequencyLowCut Pooling



FrequencyLowCut Pooling



• Aliasing and Spectral Artifact free Pooling



• Aliasing and Spectral Artifact free Pooling

For small epsilon (1/255) on ImageNet-1k
(no adversarial training!)

Method		Acc@1	Acc@5	APGD	FGSM	Corr@1
ResNet-101	Baseline	77.25	93.54	0.04	38.66	46.09
	BlurPooling 2019	78.15	94.03	0.25	42.76	46.92
	Wavelet Pooling 2020	78.06	93.96	0.04	46.11	48.49
	FLC Pooling (ours)	78.16	93.95	1.26	58.49	52.91
	ASAP _{stbl} (ours)	78.11	94.12	1.32	59.56	53.13
	ASAP _{lp} (ours)	79.07	94.35	1.57	58.99	55.82
	ASAP _{sp} (ours)	79.34	94.63	1.78	58.06	56.14

Outlook

Model robustness can be improved by understanding potential weaknesses in their building blocks.

Pooling is one such example. → Spectral properties within models

Overcompleteness might be another aspect

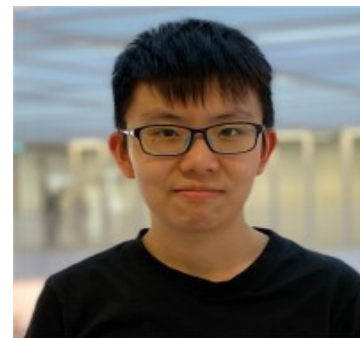
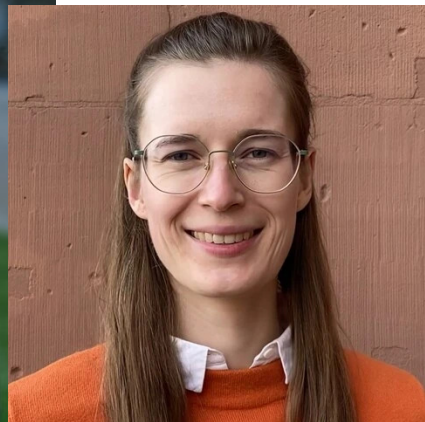
Classical rules such as: “bigger models are more robust” or “self-attention helps” don’t always apply.



10.7% a polar bear.

0.01% an arctic fox.

89.2% a seal.



Thank You!



Margret Keuper